MTH 310 **Exam 1** Name:____________________

Spring, 2026 Sections 12.1–14.3

Instructions. (100 points) Sections 12.1–14.3. You may use a calculator. Show all work for credit.

(10$^{\text{pts}}$) **1.** Let $A = (1, 3, -2)$, $B = (4, 1, 2)$, and $C = (2, 5, 0)$.

Find the scalar projection of $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

(12$^{\text{pts}}$) **2.** Consider the two lines:

$$\begin{aligned} L_1 &: \quad x = 2 + t, \quad y = 1 - t, \quad z = 3t \\ L_2 &: \quad x = 4 + 2s, \quad y = -1 - 2s, \quad z = 6s \end{aligned}$$

Are the lines **parallel**, **intersecting**, or **skew**? Justify your answer completely:

- If they are parallel, determine whether they are the **same line** or **two distinct lines**.
- If they intersect, find the point of intersection.

22 pts

(16pts) **3.** Consider the surface $4x^2 - y^2 + 4z^2 = 16$.

(a) *(8 pts)* Find the family of traces in the most important direction and at least one trace in each of the other two coordinate directions. Use these to give a complete description of the surface.

(b) *(8 pts)* Carefully sketch the surface, labeling the intercepts and the axis of symmetry.

16 pts

(16^{pts}) **4.** Let $\mathbf{r}(t) = \langle 2\cos t,\ 2\sin t,\ 3t\rangle$.

(a) *(4 pts)* Show that the curve lies on a circular cylinder and give its equation.

(b) *(7 pts)* Find the unit tangent vector $\mathbf{T}$ at $t = \pi/2$.

(c) *(5 pts)* Find parametric equations for the tangent line to the curve at $t = \pi/2$.

16 pts

(12pts) **5.** Find a vector function $\mathbf{r}(t)$ that represents the curve of intersection of the cylinder $y^2+z^2 = 4$ and the surface $x = yz$.

12 pts

(12 pts) **6.** Consider the limit:

$$\lim_{(x,y)\to(0,0)} \frac{x\,y^4}{x^4+y^4}$$

Does this limit **exist** or **not exist**? Provide careful evidence to support your conclusion. If it exists, prove it. If it does not exist, show why.

12 pts

(12pts) **7.** For $g(r, s, t) = e^r \sin(st)$, find the mixed partial derivative g_{rst}.

(10pts) **8.** The equation $xz + yz^2 = 6$ defines z implicitly as a function of x and y near the point $(1, 1, 2)$.

Find $\dfrac{\partial z}{\partial x}$ at $(1, 1, 2)$.

22 pts